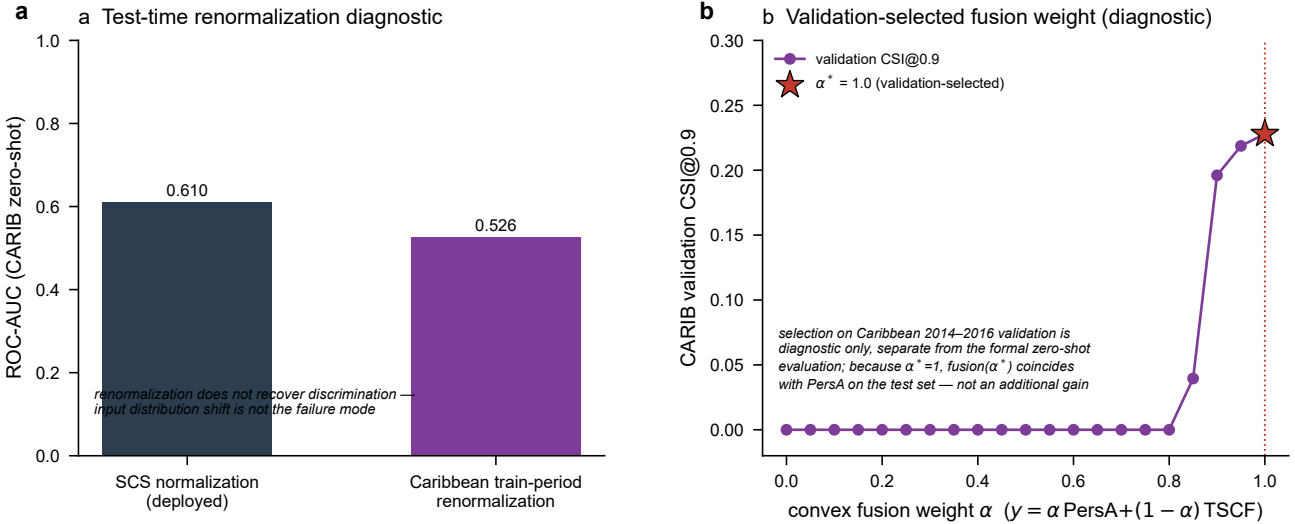

a
a Test-time renormalization diagnostic
ROC-AUC (CARIB zero-shot)
1.0
0.8
0.6
0.4
0.2
0.0
0.610
0.526
renormalization does not recover discrimination —
input distribution shift is not the failure mode
SCS normalization
(deployed)
Caribbean train-period
renormalization
b
b Validation-selected fusion weight (diagnostic)
validation CSI@0.9
α* = 1.0 (validation-selected)
CARIB validation CSI@0.9
0.30
0.25
0.20
0.15
0.10
0.05
0.00
selection on Caribbean 2014–2016 validation is
diagnostic only, separate from the formal zero-shot
evaluation; because α*=1, fusion(α*) coincides
with PersA on the test set — not an additional gain
0.0
0.2
0.4
0.6
0.8
1.0
convex fusion weight α (y = α PersA+(1 − α) TSCF)